Four AI Projects Put Structure Before Scale

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Four projects released on July 28 and 29 test the same bet: make hidden structure explicit instead of asking a larger model to infer it. CoRT compares each token's likelihood with and without rubric criteria, then redirects the usual response-level training reward. Its authors report an average +4.4% over matched GRPO runs without training a separate token scorer. DecoEvo updates a solver skill and a rubric-writing skill under separate scores, producing 2.8-5.0% relative gains over SkillOpt across five benchmarks and three model backbones. StatePlay predicts game frames alongside health, timers, and skill meters. Its mechanics-fidelity score

rose +18.6% over versions without explicit state modeling. Ai2's OlmoEarth Platform applies the same idea to infrastructure, splitting satellite preparation, GPU inference, and map assembly across hardware suited to each job.

That separation matters when one overloaded component would otherwise hide the problem. In a North America wildfire run, OlmoEarth used 19,600 CPUs and 994 GPUs to turn an estimated 4,737 serial compute hours into 30.5 wall-clock hours. The papers point to smaller, auditable changes too: targeted token credit, rubrics that expose missing criteria, and game worlds that track rules rather than only pixels.

These are promising but not direct proof that structure beats scale. The three paper results are author-reported preprints, and explicit rubrics or game-state labels add work. OlmoEarth's 155x figure compares massive parallel execution with estimated serial compute, not equal-cost systems. Still, each team found a bottleneck and represented it directly.

Read More: A survey maps more than 400 papers on agentic world models.

Sources:

- CoRT paper
- DecoEvo paper
- StatePlay paper
- Ai2: The OlmoEarth Platform

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